

The Prime Field Through One Instrument: a Ten-Row Phenomena Ledger

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Abstract

We run ten named prime-distribution phenomena through one computational instrument — the explicit formula read as prime waves, together with per-place admissibility counting — and publish the results as a ledger. The organising observation is a factorisation: nine of the ten phenomena split into a *local layer* (an all-finite-places product: decidable, classical, verified here numerically) and a *global layer* (a statement about the zero distribution of an explicit L -function, reached through the explicit formula — open in every conjectural case, and open in structurally parallel ways); the tenth (Mersenne primes) is included precisely because the factorisation *fails* there, and the ledger marks its own boundary rather than stretching past it. Headline verifications: the twin-prime count below 5×10^7 matches the Hardy–Littlewood all-places integral to 0.003%; eight Polignac gap ratios match their local products to $\leq 0.25\%$ (gap $210 \rightarrow 3.1994$ vs $16/5$); two hundred Goldbach counts match per- N predictions to 0.65% mean; the first Chebyshev lead change reproduces Leech’s $x = 26,861$ exactly; the maximal gap below 5×10^7 reproduces the record (220, 47,326,693); and three theorem-grade anchor rows (Dirichlet, Chebotarev in $\mathbb{Q}(\sqrt{5})$, Sato–Tate for the icosian cuspidal L -function) calibrate the instrument against proven mathematics, the last doubling as an audit of our own geometric Brandt eigenvalues (663 angles, all moments within pre-set 3σ gates). **Layer-1 results are classical — verified, not discovered. Nothing here proves RH, any GRH, or any infinitude conjecture.** The ledger’s sole novel claim is organisational: one local motif, one instrument, and the open parts located — each in the zero distribution of a named L -function.

1 The thesis, and what is not claimed

Fix the convention of [13, 14]: a “closure” condition at a finite place p is a congruence-admissibility condition, and the *local product* of a pattern is the product over all finite places of its admissible-density ratios. The ledger tests one statement:

Factorisation thesis. Every prime-distribution phenomenon below (row 9, the marked boundary case, excepted) factors as (i) a local layer: an all-finite-places

*Code, gates, and machine-readable artifacts: github.com/vfd-org/closure-positivity-lab, [lab/prime_ledger/](https://github.com/vfd-org/prime-ledger), [out/prime_ledger.json](https://github.com/vfd-org/prime-ledger/blob/main/out/prime_ledger.json). Work order WO-VFD-PRIME-LEDGER-001.

product, decidable and exactly computable, which fixes *local admissibility* and the singular-series constant — conjecturally, the density; and (ii) a global layer: a statement about correlations among zeros of the relevant L -functions, reached through the explicit formula, which fixes *infinitude and error terms* and is open in every conjectural row.

Scope. The local results verified here are classical — Hardy–Littlewood [1], Dirichlet, Chebotarev, Barnet-Lamb–Gee–Geraghty [7] — and the ledger claims no priority over any of them. The global statements (twin-prime and Polignac infinitude, binary Goldbach, Cramér’s conjecture, full GUE, RH itself) remain open and are not claimed, individually or in aggregate. The contribution is the uniform decomposition, the pre-registered gates, and the single reproducible artifact.

Three design rules make the ledger an instrument rather than an advertisement. *Gates are set before the data are inspected* (for statistical rows, at 3σ under the null; for deterministic rows, on exact literature constants). *Anchor rows* — phenomena that are theorems — run alongside conjectural rows on the same code path; the instrument must reproduce proven mathematics exactly or stand indicted. *Every row carries a falsifier*, stored in the machine-readable artifact.

2 The rows

All computations: sieve to $X = 5 \times 10^7$, native Python/NumPy; ζ -zeros to height 2500 by PARI/GP; total runtime ~ 11 s. One command: `python3 -m lab.prime_ledger`.

2.1 Multiplicative patterns (rows 1–3)

Row 1, twin primes. The pattern $\{0, 2\}$ is admissible at every finite place; the local product is $2C_2$ with $C_2 = \prod_{p>2} (1 - (p-1)^{-2})$, and Hardy–Littlewood’s conjecture B predicts $\pi_2(x) \sim 2C_2 \int_2^x dt / \log^2 t$. Verified: 239,101 twins counted below 5×10^7 against 239,107 predicted (ratio 0.99997), with the approach visible across checkpoints ($1.0038 \rightarrow 1.0017 \rightarrow 0.99997$). *Wall:* infinitude open; its sharpest analytic connection is to the pair correlation of zeros [11]; unconditional frontier: bounded gaps ≤ 246 [2, 3].

Row 2, Polignac gaps. The local product predicts *relative* abundances: pairs at even gap h occur $\prod_{p|h, p>2} \frac{p-1}{p-2}$ times as often as twins. Verified at eight gaps to $\leq 0.25\%$: gap 6 $\rightarrow 1.9950$ (predicted 2), gap 10 $\rightarrow 1.3318$ ($4/3$), gap 30 $\rightarrow 2.6633$ ($8/3$), gap 210 $\rightarrow 3.1994$ ($16/5$). *Wall:* same as row 1, for every fixed h .

Row 3, Goldbach. The same machinery in additive form: ordered representations $N = p+q$ track $2C_2 \prod_{p|N, p>2} \frac{p-1}{p-2} \int_2^{N-2} \frac{dt}{\log t \log(N-t)}$, a *per- N* prediction. Verified on 200 consecutive even N near 10^6 : mean deviation 0.65%, maximum 2.2%; the $3 \mid N$ doubling is conspicuous ($N = 10^6+2$: 16,400 representations vs $\sim 8,300$ – $10,800$ for its neighbours). *Wall:* positivity for every even N (binary Goldbach) open; ternary case is a theorem [4].

2.2 Equidistribution anchors (rows 4, 6, 7)

Row 4, primes in arithmetic progressions (theorem). Admissible classes mod q are exactly the residues coprime to q — the closure condition at the places dividing q . Verified: equidistribution deviation $< 0.04\%$ across $q \in \{4, 5, 12\}$; inadmissible classes exactly empty. *Wall:* the error term — square-root cancellation is GRH for Dirichlet L -functions.

Row 6, splitting in $\mathbb{Q}(\sqrt{5})$ (theorem; the programme's base field). p splits iff $p \equiv \pm 1 \pmod{5}$. Verified: split fraction 0.49993; and one *exact* closure fact — a twin pair $(p, p+2)$ with $p \equiv 3 \pmod{5}$ forces $5 \mid p+2$, so exactly one such pair exists, $(3, 5)$; the sieve finds exactly one. The Brandt prime ideals of [14] lie over precisely these split/inert primes. *Wall*: error term = zeros of $\zeta_{\mathbb{Q}(\sqrt{5})}$.

Row 7, Sato–Tate for the icosian cuspidal L (theorem [7]; self-audit). The normalized Brandt eigenvalue angles of the non-CM Hilbert modular form behind [14] must equidistribute under $(2/\pi) \sin^2 \theta$. Tested on the 663 good-prime angles (the norm-31 Steinberg prime dropped; 31 splits, and only the level ideal is removed): moments $(0.0005, 0.2515, -0.0000, 0.1265)$ against $(0, \frac{1}{4}, 0, \frac{1}{8})$, all inside pre-set 3σ gates; Kolmogorov–Smirnov distance 0.034 (5% critical value $1.36/\sqrt{663} = 0.053$). Because Sato–Tate is a *theorem* for this object, the row audits our geometric eigenvalues, not the conjecture: a persistent deviation would indict the Brandt data. *Wall*: the *rate* (effective Sato–Tate) = symmetric-power L -functions; $\sqrt{\cdot}$ -cancellation is the GRH-regime signature.

2.3 Interference rows (rows 5, 10)

Row 5, Chebyshev bias. In explicit-formula terms the bias is interference: the prime-*square* wave loads one residue class, displacing counts by $\sim \sqrt{x}/\log x$ — an asymmetric fringe on the same instrument as the prime-wave reconstruction of [14]. Verified: first mod-4 lead change at exactly $x = 26,861$ [5]; lead fraction 99.93% of primes below 5×10^7 ; mod-3 bias never reverses below the sieve bound (first reversal known near 6×10^{11}). *Wall*: the logarithmic density $0.9959\dots$ of the lead set is a theorem only under GRH + linear independence [6]; unconditionally, even existence of the density is open.

Row 10, zero statistics — the central wall, measured. The global object behind rows 1, 2 and 8 — ζ 's own zero statistics, the prototype for the parallel Dirichlet and symmetric-power walls of the other rows — is measured directly on 1,983 ζ -zeros to height 2500: unfolded spacing mean 1.0011; level repulsion $P(s < 0.2) = 0.0025$ where Poisson statistics would give 0.181 — a $70\times$ suppression, the decisive qualitative GUE signature at this height; spacing variance 0.1502 and pair-correlation deviation 0.094 from $1 - \text{sinc}^2$, both inside the documented finite-height correction scale $1/\log(T/2\pi) \approx 0.167$ [10, 9]. Montgomery's theorem [8] covers restricted test functions only; the full GUE law is conjectural. This row has no deeper layer: proving these correlations — or RH, which they refine — *is* the prototype of the walls the other rows point at.

2.4 Extremes and the boundary (rows 8, 9)

Row 8, Cramér gaps. Theorem-grade part verified: mean gap equals $\log x$ to $\leq 0.12\%$ in three windows; deterministic part: the maximal gap below 5×10^7 is exactly 220, after 47,326,693, reproducing the record tables; Cramér ratio $220/\log^2(47,326,693) = 0.704$. The pure exponential model's tail is reported *with* its known defect: $P(u > 2) = 0.113$ vs the model's 0.135 — the Hardy–Littlewood correlations of rows 1–2 are exactly the corrections the random model misses, a cross-row consistency check. *Wall*: Cramér's $O(\log^2 p)$ open; RH gives only $O(\sqrt{p} \log p)$; unconditional $\ll x^{0.525}$ [12].

Row 9, Mersenne primes — the honest boundary. Shape-constrained primes have *no L -function handle*: the instrument does not apply, and the ledger says so rather than stretching. What survives is rigid local structure (every factor q of $2^p - 1$ obeys $q \equiv 1 \pmod{2p}$ and $q \equiv \pm 1 \pmod{8}$ — verified on the actual smallest factors) and a count heuristic (Lenstra–

Pomerance–Wagstaff $\approx (e^\gamma / \log 2) \log y$: predicts 20.8 ± 4.6 below $y = 3300$; observed 18) that is *not* a singular-series consequence. Deterministic anchor: a from-scratch Lucas–Lehmer sweep reproduces the literature’s eighteen exponents exactly. *Wall, doubly open*: infinitude of Mersenne primes *and* of Mersenne composites.

3 The ledger

row	phenomenon	local layer verified	global wall (open unless noted)
1	twin primes	0.003% vs HL integral	pair corr. of zeros
2	Polignac gaps	8 ratios $\leq 0.25\%$	same, each fixed h
3	Goldbach	200 N , mean 0.65%	minor arcs / Dirichlet zeros
4	primes in APs	$< 0.04\%$; exact emptiness	error term = GRH (<i>law: theorem</i>)
5	Chebyshev bias	Leech’s 26,861 exact	density needs GRH+LI
6	splitting in $\mathbb{Q}(\sqrt{5})$	0.49993; (3, 5) unique	error = $\zeta_{\mathbb{Q}(\sqrt{5})}$ zeros (<i>law: theorem</i>)
7	Sato–Tate (our L)	663 angles, 3σ	rate = sym-power L_s (<i>law: theorem</i>)
8	Cramér gaps	mean exact; record (220)	$O(\log^2 p)$ vs $O(\sqrt{p} \log p)$
9	Mersenne	factors exact; LL 18/18	beyond the instrument
10	zero statistics	GUE at height scale	measures ζ ’s wall directly

Three rows are theorems (4, 6, 7): they are the calibration. One row (9) is the deliberate boundary marker. Each of the remaining six has its open part in the zero distribution of an explicit L -function — ζ ’s zero correlations for rows 1, 2, 8 and 10; Dirichlet L -functions for rows 3 and 5; a family of parallel walls, not literally one. That parallel structure — checkable row by row, falsifiable row by row — is the ledger’s entire claim.

Falsifier for the thesis itself. A prime-distribution phenomenon whose local layer cannot be written as an all-finite-places product, or whose global layer does not reduce to zero-correlations through the explicit formula, breaks the factorisation thesis; by the ledger’s rules it would be published as a counterexample row, not omitted.

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